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Exponential fertilization promotes seedling growth by increasing nitrogen retranslocation in trembling aspen planted for oil sands reclamation

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ABSTRACT

Vegetation reestablishment in land reclamation is often challenged by high mortality and slow growth of planted species because of low nutrient availability and severe understory competition. We tested the effectiveness of exponential fertilization in nursery trembling aspen (*Populus tremuloides* Michx.) seedling production for improving revegetation success on reconstructed soils in the Oil Sands in a two-year field experiment using a 2 (exponentially vs conventionally fertilized seedlings) × 2 (weed intact vs weed removed) factorial design. The experiment was conducted on two different cover soil types: peat mineral soil mix (PMM) and LFH mineral soil mix (LFH). Nitrogen (N) retranslocation in outplanted seedlings was traced using ^{15}N labeling. Exponential fertilization and weed removal increased height and root collar diameter growth but did not affect seedling survival over two growing seasons. Exponential fertilization increased new stem and leaf biomass and N content but decreased the percent allocation of biomass to roots. On average, 80% (on the PMM site) and 73% (on the LFH site) of total N demand of new tissues was met by internal N retranslocation. Exponential fertilization increased N retranslocation by 34% ($P < 0.01$) and 25% ($P = 0.02$) on the PMM and LFH sites, respectively. Weed competition reduced N retranslocation by 37% ($P < 0.01$) and N uptake by seedlings from the soil by 61% ($P = 0.01$) on the LFH site. We conclude that greater accumulation of nutrient reserves and greater N retranslocation helped to increase the growth of exponentially fertilized aspen seedlings that were outplanted for oil sands reclamation. Exponential fertilization of aspen seedlings for oil sands reclamation should be operationally tested for improving land reclamation in the Oil Sands.

1. Introduction

Oil sands extraction by open-pit mining in northern Alberta has disturbed a large area of the boreal region (about 813 km^2 as of December, 2013; Government of Alberta, 2014), and reclamation of these disturbed lands to an equivalent capability similar to that which existed pre-disturbance has been a priority for the industry and the provincial government. During open pit mining all vegetation, surface soil, overburden (nutrient poor layers of rock and soil below the surface soil) and oil sands are removed, leaving large pits (several kilometers wide and up to 100 m deep) or the unwanted mine dump where overburden from the mine is placed. A soil is reconstructed during reclamation of these pits and overburden storage areas after mineral extraction is complete using tailings sand (a by-product of bitumen extraction) or overburden substrates. Reclamation cover soils are then placed on these substrates to complete soil reconstruction depending on the substrate type and landform. Experience with revegetation in such reconstructed soils in reclamation of upland sites is increasing (Macdonald et al., 2015). However, success of revegetation in a reclaimed area largely depends on the establishment of understory vegetation and planted tree seedlings (Kimaro and Salifu, 2011) as most of the disturbed area was originally boreal forest. Reclamation success is often affected by unsuitable growth conditions, such as nitrogen (N) deficiency and soil salinity with electrical conductivity greater than 4 ds m^{-1} (Duan et al., 2015), pH (Jamro et al., 2014; Zhang et al., 2013), and soil compaction (Jamro et al., 2015) that is associated with reconstructed soils in oil sands reclamation. Although the use of peat mineral soil mix (PMM) or LFH (organic horizons that sit on the surface of a forest soil) mineral soil mix (LFH) as capping materials placed over tailings sand or overburden substrates has been a common practice for improving soil quality

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(Kwak et al., 2015; MacKenzie and Naeth, 2007; Rowland et al., 2009), early establishment of planted seedlings can still be problematic (Duan et al., 2015; Landhaeusser et al., 2012). In addition to the specific physical/chemical site conditions described above, cover crops that are sometimes established on these reclaimed soils to control soil erosion and may increase the competition with planted seedlings for resources such as light, water, and nutrients (Franklin et al., 2012). Field fertilization could alleviate nutrient deficiencies (Duan et al., 2015; Rowland et al., 2009), but may also encourage the growth of the understory vegetation, which subsequently intensifies competition for nutrients and other resources (Chang et al., 1996).

An alternative approach to improving revegetation success in heavily disturbed and competitive sites involves the use of quality seedling stocks (Landhaeusser et al., 2012). The nutrient loading technique based on exponential fertilization (Timmer and Aidelbaum, 1996) is designed to increase the nutrient reserve in seedling stocks by inducing luxury nutrient consumption in nursery seedling production (Malik and Timmer, 1998); this is one of the techniques that can be employed to produce quality seedling stocks. Nutrient-loaded seedlings have access to a larger nutrient reserve and can promote early seedling establishment in the field, resulting in superior survival, growth, and competitiveness over conventionally produced seedlings (Timmer and Aidelbaum, 1996; Timmer and Munson, 1991). Exponential nutrient fertilization has been successfully used to improve early establishment of seedlings of coniferous species such as black spruce (*Picea mariana* [Mill.] BSP) (Imo and Timmer, 2001; Malik and Timmer, 1996; Salifu and Timmer, 2003b), China-fir (*Cunninghamia lanceolata* [Lamb.] Hook.) (Xu and Timmer, 1999), and Lutz spruce (*Picea* × *lutzii* Little) (Jonsdottir et al., 2013). Outplanting experience with nutrient loaded deciduous species is limited (Birge et al., 2006), and non-existent with trembling aspen (*Populus tremuloides* Michx.), which is a widely distributed species across boreal North America, and is commonly used for land reclamation and commercial forestry (Pinno et al., 2012). Moreover, successful application of nutrient loading in aspen seedlings has not been tested in oil sands reclamation. However, nursery or greenhouse pot studies have been conducted on aspen (Hu, 2012; Hu et al., 2015; Schott et al., 2013). In the early stage after outplanting, seedlings can meet part of the sink demand for N that is required for new growth by retranslocating nutrients from old tissues; this has been demonstrated especially with evergreen species (Nambiar and Fife, 1991; Salifu and Timmer, 2003a). Quantification of N retranslocation from old to new tissues, which is referred to as N derived from plants (NDFP), versus N derived (or taken up) from the soil (NDFS) in newly transplanted seedlings are very important (Choi et al., 2005) for understanding which N pool is critical for promoting seedling growth (Salifu et al., 2009a). In this study, we determined the amount of N that was retranslocated to meet the sink demand in aspen seedlings after transplantation by using the ^{15}N tracing technique, which can more precisely discriminate the various N sources (such as NDFP) in the soil-plant system (Barraclough, 1995; Nommik, 1990; Proe and Millard, 1994). Although several studies have quantified NDFP, which accounts for 40–100% of annual N demand in newly transplanted conifer seedlings (Millard, 1996; Nambiar and Fife, 1991; Salifu and Timmer, 2003a), the relative contributions made by NDFP and NDFS to new tissue growth in nutrient-loaded hardwood seedlings are poorly understood (Salifu et al., 2009a).

To demonstrate the potential use of nutrient-loaded seedlings of aspen for improving revegetation success in heavily disturbed sites, we studied the growth of aspen and N retranslocation within its tissues on newly reclaimed sites with both PMM and LFH as cover soils. The field experiment was conducted in the Oil Sands region of northern Alberta. The objectives of this study were to examine the growth performance and to determine N retranslocation response of exponentially and conventionally fertilized seedlings in competition with understory vegetation on reconstructed soils. In this study we tested the following hypotheses: (i) exponentially fertilized aspen seedlings would have better growth than conventionally fertilized ones after transplantation due to the greater N storage in the exponentially fertilized seedlings, (ii) there would be a greater contribution of NDFP to new tissue growth in the exponentially than in the conventionally fertilized seedlings because of the greater nutrient reserve in the former, and (iii) the effect of understory vegetation competition for nutrients would be lower with the exponentially fertilized than with the conventionally fertilized seedlings again due to greater nutrient retranslocation to new growth in the nutrient-loaded seedlings as weed competition will not directly affect nutrient retranslocation. We focused on N in this study because N is usually the growth limiting nutrient in reclaimed soils in the Oil Sands (Duan et al., 2015; Rowland et al., 2009). Its internal cycling is similar to other mobile macronutrients (Nambiar and Fife, 1991). Thus, an improved understanding on the N nutrition has implications for the management of other mobile macronutrients.

2. Materials and methods

2.1. Seedling nursery production and fertilization regimes

Containerized stocks of conventionally (C) and exponentially fertilized (E) trembling aspen seedlings were produced in 2013 at the Smoky Lake Forest Nursery (Smoky Lake, AB, Canada). In the conventional fertilization regime, a seasonal total of 120 mg N per seedling was applied for twelve weeks at a constant rate delivered weekly; this rate was based on the rate being used in the Smoky Lake Nursery that had been found to be the optimum application rate. In the exponential fertilization regime, 240 mg N per seedling (the optimum exponential fertilization rate identified in the nursery part of the study (Hu, 2012)) was delivered on a schedule based on the modified exponential model (Birge et al., 2006; Imo and Timmer, 1992). Details of the nursery culture of seedlings can be found in Hu (2012) and Hu et al. (2015). To determine NDFP and NDFS in seedlings after transplantation, labeled urea (60 atom%) was applied in solution to the seedlings in week eight during nursery seedling production. After twelve weeks of nursery culture, the seedlings were tested for hardiness, harvested, and kept in a cold storage (at −2 °C) until they could be taken to the field for transplanting.

Before transplanting, the nutritional status, size, and biomass of seedlings were determined to assess whether exponentially fertilized seedlings were successfully loaded or not with the required nutrients. To do that, three seedling samples of each of exponential and conventional fertilization from each of five storage boxes representing different replications used in nursery production were collected from cold storage just before transplantation. Individual seedlings were assessed for component biomass and nutrient concentrations. Nutrient concentrations in the seedlings were analyzed following the procedure described below. Exponentially fertilized seedlings were confirmed to have a size and component biomass similar to that of conventionally produced seedlings but different N status prior to transplanting (Table 1).

2.2. Study site

The study was conducted at a reclamation site (56°58′N and 111°19′W) near Fort McMurray, in northern Alberta. The area was characterized by long cold winters and short warm summers, with mean annual temperature of 1 °C and precipitation of 418.6 mm (316.3 mm as rain and 102.3 mm as snow) between

Table 1
Biomass and N status (mean ± SE, n = 3) of seedlings before transplanting.

Seedling component	Dry mass (g seedling^{-1})		N concentration (mg g^{-1})		N content (mg seedling^{-1})		^{15}N content (mg seedling^{-1})		Atom% ^{15}N	
	C	E	C	E	C	E	C	E	C	E
Stem	1.54 (0.17)	1.82 (0.15)	21.64 b (1.09)	27.20 a (0.33)	33.02 b (2.43)	49.66 a (4.09)	0.49 (0.02)	0.61 (0.04)	1.53 (0.14)	1.25 (0.06)
Root	1.55 (0.28)	1.72 (0.08)	24.83 b (1.22)	34.15 a (0.58)	38.26 b (5.40)	58.80 a (3.77)	0.50 (0.08)	0.64 (0.02)	1.30 (0.03)	1.09 (0.05)
Stem + root	3.09 (0.56)	3.54 (0.06)	23.03 b (0.26)	30.60 a (0.51)	71.28 b (7.70)	108.47 a (0.31)	0.99 (0.10)	1.25 (0.01)	1.41 (0.06)	1.16 (0.01)

Abbreviations: C = conventional, E = exponential.
Different letters in the same row indicate significant differences between fertilization treatments for a particular parameter at $P < 0.05$.

1981 and 2010 (Environment Canada, 2015). The study area was located in the central mixedwood natural sub-region of the boreal forest and was dominated by trembling aspen, white spruce (*Picea glauca* [Moench] Voss) and balsam poplar (*Populus balsamifera* L.) in pure as well as in mixedwood stands (Fung and Macyk, 2000) before the forest was clearcut for oil sands mining. The natural ecosystems in the region have xeric to sub-hydric Brunisols, Luvisols and Gleysols in or Organic soils in hydric conditions (Oil Sands Vegetation Reclamation Committee, 1998). Details of basic properties of natural soils in surrounding boreal forest can be found in Jung and Chang (2012).

The research was conducted on two oil sands sites that had been reclaimed in 2008. These sites have 50 cm thick PMM and LFH as capping materials (cover soil) placed on overburden substrate and are hereafter referred to as PMM and LFH sites, respectively. In 2009, both sites were seeded with barley (*Hordeum vulgare* L.) to stabilize soil and control erosion. The PMM and LFH sites have contrasting soil properties and competing understory vegetation. The PMM has higher carbon-to-nitrogen (C:N) ratio and available N but lower bulk density than the LFH (Table 2). Percent cover by understory vegetation on the LFH was almost two times greater than that on the PMM site; aboveground biomass and N content of understory vegetation were also greater on the LFH site (Table 3).

2.3. Experimental design

The study used a completely randomized block design with two levels of nursery fertilization and two levels of weed competition in a factorial experiment replicated in four blocks. The two fertilization regimes were exponential (E) and conventional fertilization (C) in nursery seedling production, and the two weed competition treatments were weed intact (control, +W) and weed removed (treated, −W). The weed removal treatment was maintained by periodic manual removal of aboveground weeds. Treatments were randomly allocated to the plots (5 × 5 m) within each block. Adjacent blocks were separated by a 2-m buffer and adjacent plots by a 1-m buffer. A total of 400 seedlings (2 fertilization regimes × 2 weed treatments × 25 seedlings per plot × 4 replications) were planted at 1 × 1 m spacing at each site. Nursery-produced seedlings were transplanted onto the study sites on 11 June 2014.

Table 3
Percent cover, aboveground biomass and N uptake (mean ± SE, $n = 4$) by understory vegetation on the peat mineral soil mix (PMM) and LFH mineral soil mix (LFH) sites.

Site	Fertilization treatment	% Cover	Biomass (g m^{-2})	N uptake (g m^{-2})
PMM	Conventional	37.2 (5.3)	245.1 (28.4)	3.25 (0.4)
	Exponential	41.6 (4.3)	237.8 (14.4)	2.88 (0.5)
LFH	Conventional	76.0 (6.4)	347.3 (49.3)	6.02 (0.6)
	Exponential	75.7 (1.9)	294.5 (18.3)	4.70 (0.5)

2.4. Seedling growth and understory vegetation assessment

The initial size of each seedling, including shoot height (37.2 ± 2.6 cm) and root collar diameter (RCD) (3.34 ± 0.4 mm), was measured at the time of transplanting to the field; final seedling size was measured at the end of the active growing season in August of both 2014 and 2015. Shoot height, RCD, and survival rate were determined for all 25 seedlings in each plot. Shoot height was measured using a meter stick (to 1 mm accuracy) and RCD was measured using a vernier caliper (to 0.05 mm accuracy) in two directions perpendicular to one another at ground level, and the values were averaged to represent the RCD of each seedling. The growth in height and RCD were calculated as the difference between final seedling size at the end of each growing season and the initial size at transplanting. For understory vegetation assessment, two 50 × 50 cm quadrats were randomly established in each plot in August 2014 and 2015. Percent cover was determined in each quadrat for all vegetation and each life form of plant species, including shrubs, forbs, grasses and mosses/lichens.

2.5. Soil and plant sampling

Soil samples were collected from two depths (0–10 and 10–30 cm) in June 2014 using an auger from five randomly selected locations in each plot and mixed to obtain a composite sample

Table 2
Chemical and physical properties (mean ± SE, $n = 16$) of peat mineral soil mix (PMM) and LFH mineral soil mix (LFH) used in the study sites.

Soil	Depth (cm)	pH	EC (ds m^{-1})	Total C (g kg^{-1})	Total N (g kg^{-1})	C:N	NH_4^+-N (mg kg^{-1})	NO_3^--N (mg kg^{-1})	Bulk density (g cm^{-3})
PMM	0–10	6.85 (0.05)	1.88 (0.08)	98.2 (8.7)	3.1 (0.4)	33.3 (1.5)	3.36 (0.34)	5.02 (0.80)	0.58 (0.07)
	10–30	7.10 (0.05)	1.86 (0.13)	109.2 (13.9)	2.7 (0.3)	41.2 (2.0)	1.78 (0.47)	2.65 (0.47)	0.45 (0.05)
LFH	0–10	6.57 (0.07)	2.02 (0.16)	80.7 (9.3)	4.9 (0.7)	17.6 (1.3)	7.35 (0.82)	8.03 (0.44)	0.77 (0.07)
	10–30	6.71 (0.06)	1.63 (0.16)	78.8 (7.5)	3.5 (0.3)	21.4 (1.7)	5.78 (0.41)	5.02 (0.73)	0.84 (0.10)

Abbreviations: EC = electrical conductivity, C:N = carbon to nitrogen ratio.

for each layer. Bulk density in the 0–10 and 10–30 cm layers was determined by sampling of soil from three randomly selected locations in each plot using a steel corer with a volume of 98.12 cm^3. Two (in the first growing season) and five seedlings (in the second growing season) were randomly selected from each plot and destructively harvested at the end of the active growing season in 2014 and 2015. The lower number of seedlings that were sampled at the end of the first growing season was to prevent too many seedlings from being lost to harvesting. The experiment was terminated at the end of the second growing season and, thus, a higher number of seedlings were sampled for analysis to improve their representativeness. At each sampling, seedlings were cut at the ground level, followed by excavation of the root system. After the plant material was brought back to the laboratory, the aboveground parts were separated into old stems, new stems, and leaves (being a deciduous species, the leaves were produced in the current growing season). The root of each seedling was washed on to a sieve (0.5 mm mesh), most roots were recovered by carefully removing soil and peat from the root system, and as many broken roots that were retained on the sieve were manually collected as possible. Aboveground understory vegetation was collected in August 2014 for the determination of its N content. Seedling component and understory vegetation samples were washed twice with distilled water and dried for 72 h at 70 °C for determining dry mass and for chemical analyses.

2.6. Soil and plant analyses

Soil samples were air-dried at room temperature and ground to pass a 2 mm sieve; a subsample was further ground to powder using a ball mill (Mixer Mill MM 200, Thomas Scientific, Swedesboro, NJ) for analysis of total C and total N concentrations. The total C and total N were analyzed by the dry combustion method using an automated elemental analyzer (NA-1500 series, Carlo Erba, Milan, Italy). Soil pH was measured in a 0.01 mol L^{-1} $CaCl_2$ solution at a 1:2 (m:v) ratio using a pH meter (Orion, Thermo Fisher Scientific Inc., Beverly, MA, USA) and electrical conductivity (EC) was measured after soil:water extraction at 1:1 (m:v) ratio using an AP75 portable waterproof conductivity/TDS meter (Thermo Fisher Scientific Inc., Waltham, MA, USA) (Kalra and Maynard, 1991). For available N (NH_4^+-N and NO_3^--N) analysis, soil samples were extracted with 2 mol L^{-1} KCl and the extract was analyzed colorimetrically by the indophenol blue method for NH_4^+-N (Keeney and Nelson, 1982) and by the vanadium oxidation method for NO_3^--N (Miranda et al., 2001).

Component dry mass was measured for each seedling while plant component samples were composited per plot for nutrient analysis. Prior to chemical analysis, plant samples were ground to powder using the ball mill described above. The N concentration and ^{15}N abundance in the plant samples were analyzed using a stable isotope ratio mass spectrometer (Thermo Delta Plus XP IRMS, Waltham, MA, USA) linked to a CN analyzer (Costech 4010, Valencia, CA, USA) at the Stable Isotope Facility of the University of Wyoming (Laramie, WY, USA).

2.7. Data analyses

New tissues of seedlings were assumed to acquire N from two sources: N remobilized from old plant tissues or NDFP and N taken up from the soil or NDFS. The NDFP in new tissues represents N retranslocation and was calculated using the following mass balance method (Hauck and Bremner, 1976):

$$\mathrm{NDFP} = \mathrm{TN}[(A - B/(C - B)]$$

where TN is the total N content in new tissues calculated as the concentration (mg g^{-1}) multiplied by the biomass of the plant tissue, A is atom% ^{15}N in new tissues, B is the atom% ^{15}N in a natural standard (0.3663), and C is the atom% ^{15}N of the plant at transplantation which was calculated as the weighted mean of atom% ^{15}N of stems and atom% ^{15}N of roots.

The NDFS was calculated as:

$$\mathrm{NDFS} = \mathrm{TN{-}NDFP}$$

Labeled N content per plant was calculated by summing the ^{15}N in different tissues.

Growth and nutritional data of seedlings before transplantation were analyzed by a one-way analysis of variance (ANOVA) and those from the field experiment by a two-way ANOVA to examine the statistical significance of a variable using the Proc MIXED model (SAS 9.2, SAS Institute, Cary NC). The linear model for the two-way ANOVA was:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \tau_k + \varepsilon_{ijk}$$

where Y_{ijk} is a dependent variable, μ is the overall mean, α_i and β_j are the fixed effects of the ith and jth fertilization and weed removal treatments, respectively, $(\alpha\beta)_{ij}$ is the fixed effect of interaction of the two treatments, τ_k is the random effect of the kth block and ε_{ijk} is the random error within the experiment; estimation of the variance-covariance matrix involved the restricted maximum likelihood (REML) method. Correlation analyses were performed for height increment (HI), RCD increment (RCDI), biomass increment (BI) and nutrient concentrations in seedling components, understory vegetation biomass and N uptake, and soil properties. Prior to ANOVA and correlation analyses, data were tested for normality of distribution (Shapiro–Wilk test) in residuals using Proc UNIVARIATE. All data were confirmed to be in normal distribution except NDFS on the PMM site. A log-transformation (base 10) was applied for the NDFS data to meet the assumption of normality. An α = 0.05 was used to determine statistical significance in all analyses, and means of the variables were separated using the least significant difference (LSD) test.

3. Results

3.1. Seedling size and biomass

There were no significant interactions between nursery fertilization and weed control on seedling size or dry matter yield at both sites on both years (Table S1). At the end of the transplantation experiment (after the 2nd growing season), exponential fertilization significantly increased seedling size on both PMM and LFH sites (Tables 4 and S1). Exponential fertilization increased shoot height on the PMM site in 2014 and 2015 ($P = 0.016$ and 0.028, respectively) and on the LFH site in 2014 ($P = 0.022$). RCD increased on the PMM site in 2014 ($P = 0.002$) and on the LFH site in 2015 ($P = 0.038$). Weed removal increased seedling RCD in both years ($P = 0.038$ and 0.009, respectively) on the LFH site but not on the PMM site (Table 4).

Both treatments significantly increased component and total biomass of seedlings at the end of first and second growing seasons (Tables 4 and S1). Exponential fertilization increased new stem, old stem, root, and total biomass in the first growing season and leaf, new stem and total biomass in the second growing season (all $P < 0.05$) on the PMM site. On the LFH site, the increase (by 34%) was in new stem biomass ($P = 0.03$) during the first growing season and leaf (by 50%, $P = 0.02$), new stem (by 69%, $P = 0.01$), old stem (by 41%, $P < 0.01$) and total biomass (by 36%, $P < 0.01$) in the second growing season. Weed removal did not change component biomass, except 79% for new stem growth ($P = 0.01$) in the second growing season on the PMM site. Weed-removed plots yielded 92% ($P < 0.01$) more leaf biomass in the first growing season, and

Table 4
Survival, height, RCD (root collar diameter) and biomass of seedling components (mean ± SE, n = 8) over two growing seasons after transplanting.

Year	Site	Treatment	Survival	Height	RCD	Leaf	New stem	Old stem	Root	Total
			(%)	(cm)	(mm)	g seedling^{-1} of biomass				
2014	PMM	Fertilization								
		C	94.0	42.33 b	4.97 b	1.4	0.52 b	2.22 b	1.96 b	6.11 b
		E	92.7	46.94 a	5.34 a	1.82	1.06 a	3.61 a	3.13 a	9.63 a
		Weed control								
		+W	92.0	46.84 a	5.12	1.62	0.77	2.65	2.25	7.29
		−W	94.7	42.44 b	5.19	1.61	0.82	3.18	2.84	8.44
	LFH	Fertilization								
		C	93.0	41.36 b	5.03	1.39	0.56 b	3.01	2.01	6.94
		E	88.0	45.41 a	5.38	1.5	0.75 a	3.82	2.54	8.62
		Weed control								
		+W	90.0	43.26	4.97 b	0.99 b	0.61	3.28	2.01	6.89
		−W	91.0	43.5	5.44 a	1.90 a	0.69	3.54	2.55	8.68
2015	PMM	Fertilization								
		C	88.0	50.85 b	6.37	1.63 a	0.65 a	4.39	4.36	11.02 a
		E	85.5	56.88 a	6.98	2.77 b	1.48 b	5.46	4.99	14.86 b
		Weed control								
		+W	87.5	55.23	6.54	1.87	0.76 a	4.85	4.44	11.96
		−W	86.0	52.5	6.81	2.53	1.36 b	5.01	4.91	13.55
	LFH	Fertilization								
		C	82.0	49.03	6.21 b	1.95 a	1.01 a	4.74 a	4.33	12.05 a
		E	77.7	54.04	6.69 a	2.93 b	1.71 b	6.70 b	5.09	16.44 b
		Weed control								
		+W	81.5	48.29 b	6.12 b	1.97 a	0.98 a	5.5	4.4	12.86
		−W	78.5	54.79 a	6.77 a	2.91 b	1.74 b	5.94	5.03	15.62

Abbreviations: PMM = peat mineral soil mix, LFH = LFH mineral soil mix, C = conventional, E = exponential, +W = weed intact, −W = weed removed.
Different letters in the same column indicate significant differences between fertilization regimes and between weed control treatments in each year at $P < 0.05$.

48% ($P = 0.03$) and 77% ($P = 0.01$) more leaf and new stem biomass, respectively, in the second growing season compared to the weed-intact plots on the LFH site.

3.2. Percent biomass allocation to seedling components

Nursery fertilization and weed removal in the field significantly affected seedling component biomass partitioning (expressed as% of the total biomass of the seedling) at the end of the second growing season (Fig. 1 and Table S2). Percent root biomass allocation was not affected by nursery fertilization or weed competition treatments on the PMM site (Fig. 1a), but was greater in the C + W than in the other treatment combinations on the LFH site (Fig. 1b). New stem biomass allocation was significantly increased by both exponential fertilization ($P < 0.01$) and weed removal ($P < 0.01$) on the PMM site and by weed removal ($P = 0.03$) on the LFH site. Although not significant, exponential fertilization also tended to increase percent leaf biomass on both the PMM and LFH sites ($P = 0.21$ and 0.05, respectively).

3.3. Nitrogen status of seedlings

The N concentrations of seedlings at the end of the first growing season did not differ among the treatments, except leaves on the PMM site, but were lower than those prior to outplanting in both exponentially and conventionally fertilized seedlings (Table 5). Exponential fertilization increased leaf N concentration by 23% ($P < 0.01$) on the PMM site. Although not significant ($P = 0.24$), weed competition on the LFH site reduced leaf N concentration by 14%.

In contrast to N concentration, N content in the seedlings was significantly increased by exponential fertilization on both the PMM and LFH sites (Table 5) and by weed removal only on the LFH site. Exponential fertilization increased N content in leaves by 37% ($P = 0.01$) and N content in the seedlings by 36% ($P = 0.01$), compared to conventional fertilization on the PMM site and 5% ($P = 0.74$) and 23% ($P = 0.04$), respectively, on the LFH site. Weed competition decreased N content in leaves by 55% ($P < 0.01$) and N content in seedlings by 33% ($P = 0.01$) on the LFH site.

3.4. Relationship between seedling growth and N status, weed competition, and soil properties

RCD increment (RCDI) was correlated with foliar N concentration on the PMM site ($P = 0.04$). On the LFH site, RCDI was positively correlated with soil NH_4^+-N ($P = 0.04$), soil total C ($P = 0.01$), soil total N ($P = 0.03$), foliar N concentration ($P < 0.01$) and content ($P < 0.01$), and shoot:root ratio (S:R) ($P = 0.03$), and negatively correlated with understory aboveground biomass ($P = 0.03$) and understory aboveground N content ($P = 0.04$). The biomass increment (BI) was positively correlated with soil NH_4^+-N ($P = 0.02$), foliar N concentration ($P < 0.01$), and foliar N content ($P = 0.02$), new stems ($P = 0.04$) and old stems ($P < 0.01$) (Table S3) on the LFH site.

3.5. Nitrogen retranslocation and soil N uptake by seedlings

In comparison to N remobilized from old tissues, N uptake from the soil by aspen seedlings was much lower in the first growing season. On average, 80% (on the PMM site) and 73% (on the LFH site) of total N in new tissues was derived from old tissues during the first growing season (Fig. 2). Although percent N retranslocation was not different among the treatments, total NDFP was significantly increased by both exponential fertilization and weed removal without a significant interaction among the treatments. In new tissues, NDFP in the exponentially fertilized seedling was 25 ($P = 0.02$) and 37% ($P < 0.01$) greater than that in conventionally fertilized seedlings on the PMM and LFH sites, respectively, while weed competition reduced NDFP by 37% ($P < 0.01$) on the LFH site. In contrast to NDFP, NDFS was increased only by weed removal on the LFH site but was not affected by the fertilization treatment on both sites. The increase in NDFS by weed removal was 158% ($P = 0.01$) on the LFH site.

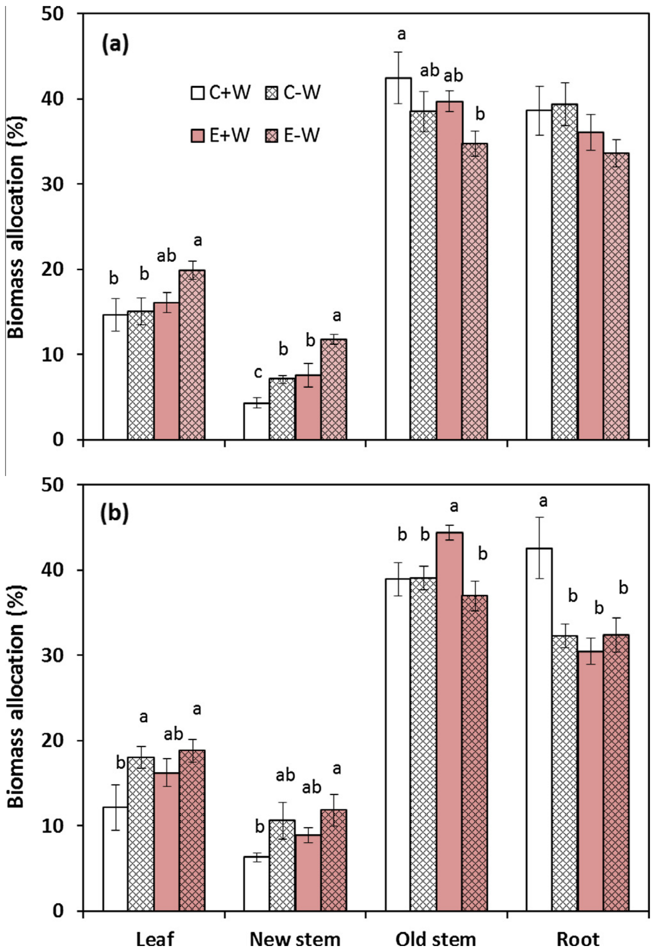

Fig. 1. Percent biomass allocation to components of seedlings on (a) the PMM (peat mineral soil mix) and (b) the LFH (LFH mineral soil mix) sites in the second growing season. Different letters represent significant differences among treatments within each seedling component at $P < 0.05$. Vertical bars are standard errors of the means ($n = 4$).

4. Discussion

4.1. Exponential fertilization and weed removal increased aspen seedling growth

Exponential fertilization in nursery seedling production stimulated the early growth of aspen seedlings after they were outplanted, supporting our first hypothesis; this indicates that exponential fertilization can help facilitate revegetation in reclaimed soils with competing vegetation, as has been reported for other site conditions (Oliet et al., 2009; Salifu et al., 2009b). In a greenhouse experiment, Hu et al. (2015) also reported increased growth of aspen seedlings as a result of nutrient loading prior to planting in reclamation soil materials. The improved growth could be attributed to the nutrient reserve that was built up in exponentially fertilized seedlings (Hu et al., 2015; Timmer and Aidelbaum, 1996), as exponential fertilization in the nursery increased N content in stems and roots and built up the nutrient reserve in the seedlings prior to transplantation.

Although total biomass was significantly greater in exponentially fertilized seedlings, percent biomass allocation to roots was greater in conventionally fertilized seedlings two growing seasons after transplantation, suggesting nutrient stress in the reclaimed soil, as nutrient stress favors root growth that would increase the seedling's ability to exploit a larger volume of soil (Nambiar and Sands, 1993); plants investment of more photosynthate in root growth is an adaptation for increasing N uptake from the soil (Ingestad and Agren, 1988). In contrast, greater allocation of biomass and N to metabolically active tissues such as leaves and new stems in the exponential fertilization and weed removal treatment combination promotes carbon fixation by increasing the surface area and chlorophyll concentration of leaves (Gulmon and Chu, 1981), which in turn supports better seedling growth. Competition from the understory vegetation was greater on the LFH than on the PMM site as reflected by the higher percent cover and aboveground biomass of and N uptake by the understory vegetation (Table 2). Under field conditions, aspen seedlings exhibited a greater response to nursery fertilization than to weed management on the PMM site, while they responded to both treatments on the LFH site. Weed removal increased shoot height and seedling biomass on the LFH site by reducing the competition for resources such as light, moisture and nutrients. In our study, understory vegetation was cut short around seedlings in weed-intact plots to eliminate light competition; soil moisture content was not significantly different between weed intact and weed-removed plots (data not shown). Increased nutrient availability that was incurred by removing weeds likely contributed to better seedling growth. Similar results regarding nutrient competition were observed by Imo and Timmer (2001) in black spruce in highly competitive forest sites. The negative effect of weed competition on seedling growth is well known and has been reported in many previous studies for coniferous species such as Monterey pine (*Pinus radiata* D. Don) (Woods et al., 1992), western red cedar (*Thuja plicata* Donn ex D. Don) (Chang et al., 1996), and white spruce (Matsushima and Chang, 2007).

The effect of the fertilization treatment on N concentration disappeared at the end of the first growing season, despite the significant difference in N concentration in seedling components before outplanting. The greater reduction in N concentration in the exponentially fertilized seedlings in the field was caused by the dilution effect that was incurred by higher biomass production in these seedlings, suggesting short-term benefits of nursery nutrient loading in terms of N concentration, however the benefits of nutrient loading in biomass production were maintained during field growth. The greater reduction in N concentration in exponentially fertilized seedlings is consistent with the findings of Jonsdottir et al. (2013) for Lutz spruce and Heiskanen et al. (2009) for Norway spruce (*Picea abies* [L.] Karsten). Although foliar N concentration in exponentially fertilized seedlings was higher than that in conventionally fertilized ones on the PMM site, foliar N concentrations in all treatments were below the optimal level (30–40 mg g^{-1}) for aspen growth (Hansen, 1994); therefore, supplementary fertilization may be required in reclaimed soils (Sloan and Jacobs, 2013), particularly after the soil nutrient availability cannot meet the demand of the outplanted trees as the trees grow older where nutrient limitation can mean the complete failure of the revegetation or reclamation effort (Duan et al., 2015). The significantly positive relationship between seedling growth and available soil N implies that the reduced availability of soil N in recently reclaimed and highly competitive LFH sites was limiting seedling growth. Furthermore, the significant correlation between growth and foliar N in seedlings on the LFH site further suggests the relevance of weed removal treatment in increasing foliar N and growth of aspen seedlings on the reclaimed soil.

4.2. Exponential fertilization and weed removal increased NDFP and NDFS

The high rate of N retranslocation (80% and 73% of total sink demand of new tissues on the PMM and LFH sites, respectively) in aspen is similar to that reported for black walnut (*Juglans nigra* L.)

Table 5
Effects of nursery fertilization and field weed control treatments on total N concentration (mg g^{-1}) and N content (mg seedling^{-1}) in seedlings (mean ± SE, n = 8) post-transplanting.

Site	Treatment	New leaf		New stem		Old stem		Root		Total	
		TN	N content	TN	N content	TN	N content	TN	N content	TN	N content
PMM	Fertilizer										
	C	11.49 b	16.10 b	10.45	5.43	5.26	10.30 b	6.38	14.59 b	7.98	46.42 b
	E	14.17 a	25.74 a	8.02	8.44	5.11	19.32 a	6.36	19.18 a	7.56	72.69 a
	Weed control										
	+W	12.85	21.37	10.23	7.46	5.42	13.20	6.81	15.12	8.21	57.15
	−W	12.83	20.47	8.24	6.42	4.94	16.41	5.91	18.66	7.33	61.95
LFH	Fertilizer										
	C	13.18	19.47	9.72	5.47 b	6.48	19.24 b	8.38	18.37 b	9.81	62.73 b
	E	13.77	20.5	11.48	8.29 a	8.16	29.77 a	9.44	23.55 a	8.82	82.11 a
	Weed control										
	+W	12.48	12.41 b	10.04	5.95	7.11	23.22	7.83	16.45	8.51	58.04 b
	−W	14.47	27.55 a	11.15	7.81	7.53	25.96	9.99	25.46	10.11	86.79 a

Abbreviations: PMM = peat mineral soil mix, LFH = LFH mineral soil mix, TN = nitrogen concentration, C = conventional, E = exponential, +W = weed intact, −W = weed removed.
Different letters in the same column indicate significant differences between fertilization regimes and between weed control treatments at P < 0.05.

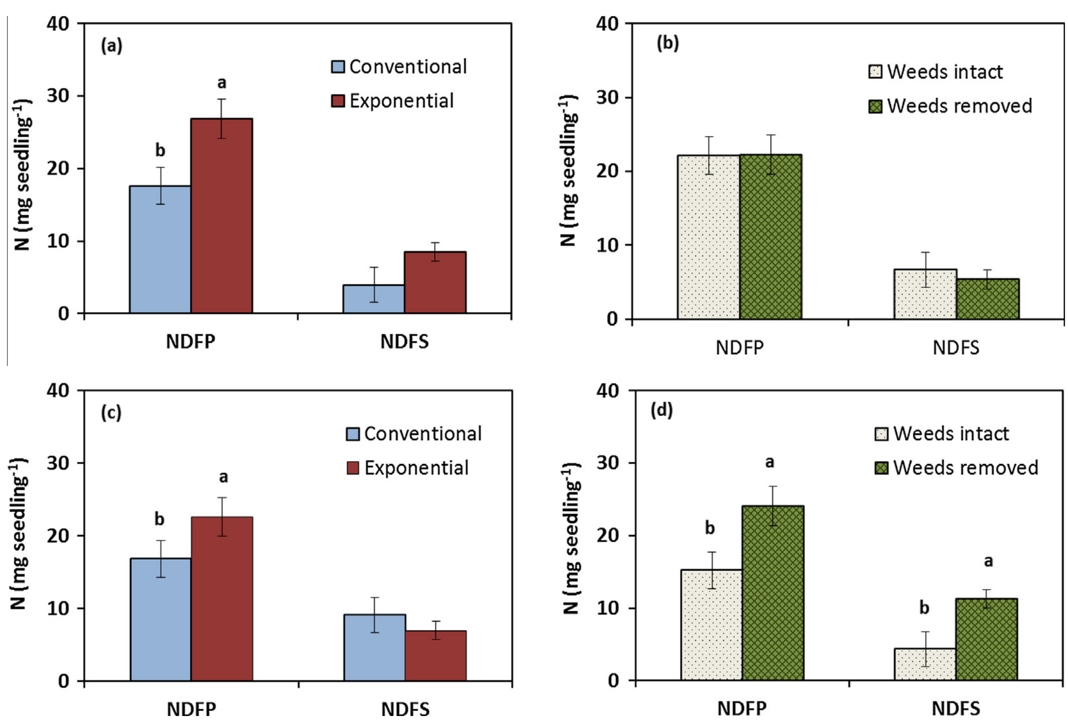

Fig. 2. Effects of nursery fertilization and weed control after field outplanting on (a) nitrogen derived from plant (NDFP) and nitrogen derived from the soil (NDFS) in seedlings on the PMM site as affected by nursery fertilization, (b) NDFP and NDFS in seedlings on the PMM site as affected by weed competition, (c) NDFP and NDFS in seedlings on the LFH site as affected by nursery fertilization, and (d) NDFP and NDFS in seedlings on the LFH site as affected by weed competition. Different letters represent significant differences between the nursery fertilization or weed control treatments at P < 0.05. Vertical bars are standard errors of the means (n = 8).

(Salifu et al., 2009a), in which NDFP accounted for 68–83% of the total N demand. However, only 32% of total N demand by new tissue was met by NDFP in northern red oak (*Quercus alba* L.) (Salifu et al., 2008), indicating that hardwood species varied widely in terms of the contribution of NDFP in meeting the total sink demand. The greater contribution of NDFP compared to NDFS in aspen seedlings suggests the importance of internal nutrient cycling to meet sink demand of new growth in early establishment after outplanting.

The increased NDFP in exponentially fertilized seedlings, owing to the greater N reserve in the seedlings prior to outplanting, supports our second hypothesis. In our study, exponential fertilization increased nutrient reserves without increasing seedling size in nursery production (Hu, 2012), suggesting that N retranslocation during field growth may be determined by the nutrient reserve but not by seedling size. The difference in net N retranslocation between exponentially and conventionally fertilized seedlings was likely determined by the amount of readily available N in the seedlings. Exponentially fertilized seedlings may have larger pools of less diluted, more readily available and mobile nutrients in the form of non-functional amino acids and proteins as a result of luxury consumption of nutrients (Malik and Timmer, 1998; McAlister and Timmer, 1998). Low N uptake from the soil compared to N remobilized in the early stage of seedling establishment after outplanting may be caused by restricted root placement and poor root-soil contact, which limits water and nutrient uptake from the soil (Burdett, 1990), or by salinity (Frota and Tucker, 1978) that contributes to reduced permeability of roots, and which subsequently decreases nutrient uptake in a reclaimed soil. The increase in N uptake from the soil on the LFH site by weed removal was linked with increased soil-available N with reduced understory competition (Woods et al., 1992).

The greater NDFP during field growth of exponentially fertilized seedlings was linked to the increase in plant biomass production, especially in new tissues. The availability of N for remobilization was more important than N uptake from the soil for the formation of new tissues in aspen seedlings in their early establishment. Growth had been reported to be sustained by internal reserves in the plant rather than by N uptake from the soil in early years of outplanting for several plant species (Millard, 1996; Millard and Neilsen, 1989; Munson et al., 1995). The fact that the seedlings having higher growth were associated with greater N retranslocation in exponentially fertilized seedlings suggests that current growth was strongly affected by N retranslocation in the planted aspen seedlings.

Nitrogen retranslocation in seedlings growing in less stressed environments (i.e., weed removal plots on the LFH site) was greater than those growing in the stressed environment (i.e., weed-intact plots). Greater N uptake from the soil was linked to the higher growth of seedlings in weed-removed plots. Presumably, stronger growth sinks caused by greater new tissue formation in aspen seedlings on weed-removed plots increased the need for internal N redistribution (Nambiar and Fife, 1991; Proe and Millard, 1994), supporting the N retranslocation driven by sink demand hypothesis. This result, however, contradicts the findings for black spruce that were found by Malik and Timmer (1998), who observed higher nutrient retranslocation in weed-intact than in weed-removed plots. They pointed out that nutrient conservation could be one of the reasons for having less N retranslocation in black spruce seedlings growing in weed-removed plots.

5. Conclusions

We conclude that N remobilization is important in meeting the nutrient requirement in the new growth in the early establishment of aspen seedlings in reclaimed soils, given that three-quarters of the demand for N by the new growth was derived from old tissues after the seedlings were transplanted. Since N retranslocation was dependent on nutrient pool size in the seedlings prior to transplantation, this study also demonstrated the significance of building up of nutrient reserves in the seedlings during nursery culture for the nutrition of seedling after transplantation. Yet the nutrient pool size before transplanting had little influence on N uptake from the soil. Exponential fertilization promoted the growth of seedlings after transplantation in the field by increasing N retranslocation, demonstrating its potential use in helping to improve revegetation success in heavily disturbed and competitive sites. The effect of understory removal was site-specific, with significant improvement in growth, N retranslocation and uptake from the soil on the LFH site (a site with more severe weed competition), but the effect was less significant on the PMM site. Our study only tested the field growth performance of exponentially fertilized seedlings of aspen for two growing seasons; the utility of the nutrient loading technique needs to be operationally tested for the longer term and applied in future land reclamation practices once operationally tested to improve the success of land reclamation using trembling aspen in the boreal region. Our two-year data did not show any nutrient-loading effect on seedling survival rate and as such nutrient-loading may not be effective in improving all aspects of seedling performance after outplanting.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.foreco.2016.03.034.

References

Barraclough, D., 1995. ^{15}N isotope dilution techniques to study soil nitrogen transformations and plant uptake. Fert. Res. 42, 185–192.

Birge, Z.K.D., Salifu, K.F., Jacobs, D.F., 2006. Modified exponential nitrogen loading to promote morphological quality and nutrient storage of bareroot-cultured *Quercus rubra* and *Quercus alba* seedlings. Scand. J. For. Res. 21, 306–316.

Burdett, A., 1990. Physiological processes in plantation establishment and the development of specifications for forest planting stock. Can. J. For. Res. 20, 415–427.

Chang, S.X., Preston, C.M., McCullough, K., Weetman, G.F., Barker, J., 1996. Effect of understory competition on distribution and recovery of ^{15}N applied to a western red cedar – western hemlock clear-cut site. Can. J. For. Res. 26, 313–321.

Choi, W.-J., Chang, S.X., Hao, X., 2005. Soil retention, tree uptake, and tree resorption of $^{15}NH_4NO_3$ and $NH_4{}^{15}NO_3$ applied to trembling and hybrid aspens at planting. Can. J. For. Res. 35, 823–831.

Duan, M., House, J., Chang, S.X., 2015. Limiting factors for lodgepole pine (*Pinus contorta*) and white spruce (*Picea glauca*) growth differ in some reconstructed sites in the Athabasca oil sands region. Ecol. Eng. 75, 323–331.

Environment Canada, 2015. Canadian climate normal 1981–2010. Meteorological Service of Canada, Environment Canada, Government of Canada. <http://www.climate.weatheroffice.ec.gc.ca/climate_normals/results_e.html> (accessed on October 26, 2015).

Franklin, J.A., Zipper, C.E., Burger, J.A., Skousen, J.G., Jacobs, D.F., 2012. Influence of herbaceous ground cover on forest restoration of eastern US coal surface mines. New For. 43, 905–924.

Frota, J., Tucker, T., 1978. Absorption rates of ammonium and nitrate by red kidney beans under salt and water stress. Soil Sci. Soc. Am. J. 42, 753–756.

Fung, M.Y.P., Macyk, T.M., 2000. Reclamation of oil sands mining areas. In: Barnhisel, R.I., Darmody, R.G., Daniels, W.L. (Eds.), Reclamation of Drastically Disturbed Lands. American Society of Agronomy, pp. 755–774.

Government of Alberta, 2014. Alberta's Oil Sands, Reclamation. <http://www.oilsands.alberta.ca/reclamation.html> (accessed on October 25, 2015).

Gulmon, S., Chu, C., 1981. The effects of light and nitrogen on photosynthesis, leaf characteristics, and dry matter allocation in the chaparral shrub, *Diplacus aurantiacus*. Oecologia 49, 207–212.

Hansen, E.A., 1994. A guide for determining when to fertilize hybrid poplar plantations. USDA Forest Service, North Central Forest Experimental Station. Research Paper NC-319 St Paul, MN.

Hauck, R., Bremner, J., 1976. Use of tracers for soil and fertilizer nitrogen research. Adv. Agron. 28, 219–266.

Heiskanen, J., Lahti, M., Luoranen, J., Rikala, R., 2009. Nutrient loading has a transitory effect on the nitrogen status and growth of outplanted Norway spruce seedlings. Silva Fenn. 43, 249–260.

Hu, Y., 2012. Nutrient loading of aspen, jack pine and white spruce seedlings for potential out-planting in oil sands reclamation (M.Sc. thesis), University of Alberta.

Hu, Y., Hu, Y., Zeng, D., Tan, X., Chang, S.X., 2015. Exponential fertilization and plant competition effects on the growth and N nutrition of trembling aspen and white spruce seedlings. Can. J. For. Res. 45, 78–86.

Imo, M., Timmer, V.R., 1992. Nitrogen uptake of mesquite seedlings at conventional and exponential fertilization schedules. Soil Sci. Soc. Am. J. 56, 927–934.

Imo, M., Timmer, V.R., 2001. Growth and nitrogen retranslocation of nutrient loaded *Picea mariana* seedlings planted on boreal mixedwood sites. Can. J. For. Res. 31, 1357–1366.

Ingestad, T., Agren, G.I., 1988. Nutrient uptake and allocation at steady-state nutrition. Physiol. Plant. 72, 450–459.

Jamro, G.M., Chang, S.X., Naeth, M.A., 2014. Organic capping type affected nitrogen availability and associated enzyme activities in reconstructed oil sands soils in Alberta, Canada. Ecol. Eng. 73, 92–101.

Jamro, G.M., Chang, S.X., Naeth, M.A., Duan, M., House, J., 2015. Fine root dynamics in lodgepole pine and white spruce stands along productivity gradients in reclaimed oil sands sites. Ecol. Evol. 5, 4655–4670.

Jonsdottir, R.J., Sigurdsson, B.D., Lindstrom, A., 2013. Effects of nutrient loading and fertilization at planting on growth and nutrient status of Lutz spruce (*Picea* $\times$ *lutzii*) seedlings during the first growing season in Iceland. Scand. J. For. Res. 28, 631–641.

Jung, K.H., Chang, S.X., 2012. Four years of simulated N and S depositions did not cause N saturation in a mixedwood boreal forest ecosystem in the oil sands region in northern Alberta, Canada. For. Ecol. Manage. 280, 62–70.

Kalra, Y.P., Maynard D.G., 1991, Methods manual for forest soil and plant analysis. Forestry Canada, Northwest Region. Northern Forestry Centre, Alberta, Canada.

Keeney, D.R., Nelson, D.W., 1982. Nitrogen-inorganic forms. In: Page, A.L., Miller, R. H., Keeny, D.R. (Eds.), Methods of Soil Analysis Part 2, Chemical and Microbiological Properties. American Society of Agronomy Inc. and Soil Science Society of America Inc., Madison, WI, pp. 643–698.

Kimaro, A., Salifu, K., 2011. The effect of cover crop species on growth and yield response of tree seedlings to fertiliser and soil moisture on reclaimed sites. In: Fourie, A.B., Tibbett, M., Bersing, A. (Eds.), Mine Closure. Australian Centre for Geomechanics, Perth, pp. 37–46.

Kwak, J., Chang, S.X., Naeth, M.A., Schaaf, W., 2015. Coarse woody debris extract decreases nitrogen availability in two reclaimed oil sands soils in Canada. Ecol. Eng. 84, 13–21.

Landhaeusser, S.M., Rodriguez-Alvarez, J., Marenholtz, E.H., Lieffers, V.J., 2012. Effect of stock type characteristics and time of planting on field performance of aspen (*Populus tremuloides* Michx.) seedlings on boreal reclamation sites. New For. 43, 679–693.

Macdonald, S.E., Landhäusser, S.M., Skousen, J., Franklin, J., Frouz, J., Hall, S., Jacobs, D.F., Quideau, S., 2015. Forest restoration following surface mining disturbance: challenges and solutions. New For. 46, 703–732.

MacKenzie, D.D., Naeth, M.A., 2007. Assisted natural recovery using a forest soil propagule bank in the Athabasca Oil Sands. In: Adkins, S.W., Asfmore, S., Navre, S.C. (Eds.), Seeds: Biology, Development and Ecology. CABI, Wallingford, United Kingdom, pp. 374–382.

Malik, V., Timmer, V.R., 1996. Growth, nutrient dynamics, and interspecific competition of nutrient-loaded black spruce seedlings on a boreal mixedwood site. Can. J. For. Res. 26, 1651–1659.

Malik, V., Timmer, V.R., 1998. Biomass partitioning and nitrogen retranslocation in black spruce seedlings on competitive mixedwood sites: a bioassay study. Can. J. For. Res. 28, 206–215.

Matsushima, M., Chang, S.X., 2007. Effects of understory removal, N fertilization, and litter layer removal on soil N cycling in a 13-year-old white spruce plantation infested with Canada bluejoint grass. Plant Soil 292, 243–258.

McAlister, J.A., Timmer, V.R., 1998. Nutrient enrichment of white spruce seedlings during nursery culture and initial plantation establishment. Tree Physiol. 18, 195–202.

Millard, P., 1996. Ecophysiology of the internal cycling of nitrogen for tree growth. Z. Pflanzen. Bodenk. 159, 1–10.

Millard, P., Neilsen, G., 1989. The influence of nitrogen supply on the uptake and remobilization of stored N for the seasonal growth of apple trees. Ann. Bot. 63, 301–309.

Miranda, K.M., Espey, M.G., Wink, D.A., 2001. A rapid, simple spectrophotometric method for simultaneous detection of nitrate and nitrite. Nitric Oxide 5, 62–71.

Munson, A.D., Margolis, H.A., Brand, D.G., 1995. Seasonal nutrient dynamics in white pine and white spruce in response to environmental manipulation. Tree Physiol. 15, 141–149.

Nambiar, E.K.S., Fife, D.N., 1991. Nutrient retranslocation in temperate conifers. Tree Physiol. 9, 185–207.

Nambiar, E.K.S., Sands, R., 1993. Competition for water and nutrients in forests. Can. J. For. Res. 23, 1955–1968.

Nommik, H., 1990. Application of ^{15}N as a tracer in studying fertiliser nitrogen transformations and recovery in coniferous ecosystems. In: Harrison, A.P., Ineson, P., Heal, O.W. (Eds.), Nutrient Cycling in Terrestrial Ecosystems: Field Methods, Application and Interpretation. Elsevier Applied Science, New York, NY, pp. 276–290.

Oil Sands Vegetation Committee, 1998. Guidelines for reclamation to forest vegetation in the Athabasca oil sands region. Report # ESD/LM/99-1. Prov. Gov. of Alberta, Edmonton, AB. <http://environment.gov.ab.ca/info/library/6869.pdf> (assessed January 21, 2016).

Oliet, J.A., Tejada, M., Salifu, K.F., Collazos, A., Jacobs, D.F., 2009. Performance and nutrient dynamics of holm oak (*Quercus ilex* L.) seedlings in relation to nursery nutrient loading and post-transplant fertility. Eur. J. For. Res. 128, 253–263.

Pinno, B.D., Landhaeusser, S.M., MacKenzie, M.D., Quideau, S.A., Chow, P.S., 2012. Trembling aspen seedling establishment, growth and response to fertilization on contrasting soils used in oil sands reclamation. Can. J. Soil Sci. 92, 143–151.

Proe, M., Millard, P., 1994. Relationships between nutrient supply, nitrogen partitioning and growth in young Sitka spruce (*Picea sitchensis*). Tree Physiol. 14, 75–88.

Rowland, S.M., Prescott, C.E., Grayston, S.J., Quideau, S.A., Bradfield, G.E., 2009. Recreating a functioning forest soil in reclaimed oil sands in northern Alberta: an approach for measuring success in ecological restoration. J. Environ. Qual. 38, 1580–1590.

Salifu, K.F., Apostol, K.G., Jacobs, D.F., Islam, M.A., 2008. Growth, physiology, and nutrient retranslocation in ^{15}N fertilized *Quercus rubra* seedlings. Ann. For. Sci. 65. http://dx.doi.org/10.1051/forest:2007073 101.

Salifu, K.F., Timmer, V.R., 2003a. Nitrogen retranslocation response of young *Picea mariana* to nitrogen-15 supply. Soil Sci. Soc. Am. J. 67, 309–317.

Salifu, K.F., Timmer, V.R., 2003b. Optimizing nitrogen loading of *Picea mariana* seedlings during nursery culture. Can. J. For. Res. 33, 1287–1294.

Salifu, K.F., Islam, M.A., Jacobs, D.F., 2009a. Retranslocation, plant, and soil recovery of nitrogen-15 applied to bareroot black walnut seedlings. Commun. Soil Sci. Plant Anal. 40, 1408–1417.

Salifu, K.F., Jacobs, D.F., Birge, Z.K.D., 2009b. Nursery nitrogen loading improves field performance of bareroot oak seedlings planted on abandoned mine lands. Restor. Ecol. 17, 339–349.

Schott, K.M., Pinno, B.D., Landhaeusser, S.M., 2013. Premature shoot growth termination allows nutrient loading of seedlings with an indeterminate growth strategy. New For. 44, 635–647.

Sloan, J.L., Jacobs, D.F., 2013. Fertilization at planting influences seedling growth and vegetative competition on a post-mining boreal reclamation site. New For. 44, 687–701.

Timmer, V., Aidelbaum, A., 1996. Manual for exponential nutrient loading of seedlings to improve outplanting performance on competitive forest sites, Technical Report No. TR-25. NODA/NFP, Ministry of Natural Resources of Ontario, Ontario.

Timmer, V., Munson, A., 1991. Site-specific growth and nutrition of planted *Picea mariana* in the Ontario Clay Belt. IV. Nitrogen loading response. Can. J. For. Res. 21, 1058–1065.

Woods, P., Nambiar, E., Smethurst, P., 1992. Effect of annual weeds on water and nitrogen availability to *Pinus radiata* trees in a young plantation. For. Ecol. Manage. 48, 145–163.

Xu, X.J., Timmer, V.R., 1999. Growth and nitrogen nutrition of Chinese fir seedlings exposed to nutrient loading and fertilization. Plant Soil 216, 83–91.

Zhang, W., Calvo-Polanco, M., Chen, Z.C., Zwiazek, J.J., 2013. Growth and physiological responses of trembling aspen (*Populus tremuloides*), white spruce (*Picea glauca*) and tamarack (*Larix laricina*) seedlings to root zone pH. Plant Soil 373, 775–786.